Unsupervised Clustering of Astronomical Objects using Colour Features and Redshift

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Introduction

The SDSS has catalogued hundreds of millions of astronomical objects, far exceeding what can be manually classified. Distinguishing stars, galaxies, and QSOs underpins most astrophysical analysis, but spectroscopic classification is expensive at scale. Unsupervised clustering on photometric data offers a cheaper alternative, automating catalogue organisation before costly follow-up observations.

We apply K-Means, DBSCAN, HDBSCAN, and GMM to 9,733 SDSS objects using engineered colour indices and log-transformed redshift. Each method targets a different structural assumption: K-Means provides an interpretable baseline; GMM fits ellipsoidal clusters via full covariance; DBSCAN finds arbitrarily-shaped clusters without a fixed count; HDBSCAN extends this to varying-density structures hierarchically.

Prior work highlights recurring limitations. Logan and Fotopoulou (2020) applied HDBSCAN to ~50,000 SDSS objects in colour space but relied on infrared WISE bands absent from optical-only datasets. Turner et al. (2018) noted that k-means assumes clusters are spherical in galaxy feature space. Tian et al. (2025) highlighted that astronomical spectral data are exponentially sparsely distributed in high-dimensional space. We compare all methods on a shared optical feature set, evaluated against ground-truth labels to identify where geometric assumptions are the binding constraint.

Dataset Overview

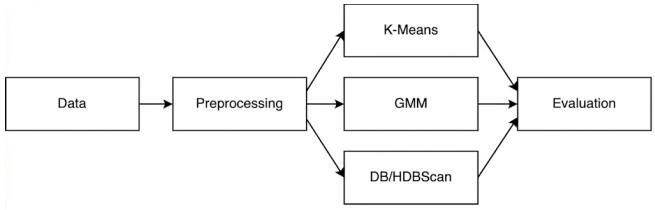
Our dataset contained 10,000 initial SDSS observations with five optical bands (u, g, r, i, z), coordinates, redshift, and spectroscopic class. After removing outliers, 9,733 observations remain with no missing values. The class distribution of the observations are: STAR (625), GALAXY (8,423), QSO (685).

Data Preprocessing

RA/Dec were dropped as they encode spatial position rather than object type. Raw band magnitudes are highly correlated (Figure 1), so we engineered four colour indices (u−g, g−r, r−i, i−z) that cancel distance effects and isolate spectral shape. Redshift is heavily right-skewed (Figure 2) so we applied a $\log_{10}$ transform to symmetrise the distribution and bring it to a comparable scale as the colour indices. Finally, RobustScaler was applied to u−g, g−r, and log-redshift, while QuantileTransformer was applied to r−i and i−z due to their extreme skewness (Figure 3). This yields five features for clustering: u−g, g−r, r−i, i−z, and $\log_{10}$(redshift) (Figure 4).

Methodology

Overall Pipeline



K-Means Clustering

K-Means is a partition-based clustering algorithm which groups data into K distinct, non-overlapping clusters by minimising the distance between the data points and their cluster centroids. K-Means was chosen due to its simplicity, efficiency and interpretability to provide crucial insight in astronomical analysis (Turner et al., 2018). The key hyperparameter is the number of clusters K, determined using the combination of the elbow plot, silhouette score and stability analysis for reproducibility. These diagnostics guided the selection of K = 2, 3, 4 as candidates for further evaluation.

Gaussian Mixture Model (GMM)

GMM adopts a model-based approach, assuming data arise from a mixture of Gaussian distributions. Unlike k-means, this allows for elliptical clusters and overlapping boundaries, which is more realistic for astronomical data where class separation is not sharp (Kuhn & Feigelson, 2017). GMM provides posterior membership probabilities, offering softer and more informative cluster assignments. The number of components was tuned using the Bayesian Information Criterion (BIC), which balances fit and complexity to prevent overfitting (Tian et al., 2025). Both K=10 (BIC selected) and K=3 (ground truth) were evaluated. BIC was also used to select the covariance structure, with “full” chosen to capture inter-feature correlations critical in

high-dimensional spectral data. Additional parameters, including multiple initialisations and sufficient EM iterations, ensured stable convergence.

DBSCAN & Hierarchical DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups points closely packed together while marking points in low-density regions as outliers. It is well-suited for datasets where clusters have arbitrary shapes, as it does not assume spherical distributions or require the number of clusters to be specified beforehand. However, DBSCAN is sensitive to its hyperparameters, particularly eps and min_samples, and can struggle when clusters have varying densities, often merging distinct clusters or misclassifying points as noise, as seen in Figure 7a. HDBSCAN extends DBSCAN by using a hierarchical clustering framework, enabling it to identify clusters of varying densities and sizes more effectively. It constructs a cluster hierarchy and extracts flat clusters based on stability, which improves the separation of overlapping clusters. The main hyperparameter is min_cluster_size, and we perform hyperparameter tuning using ranges inspired by Logan and Fotopoulou (2020). Although HDBSCAN can produce many outliers, especially with a large min_cluster_size of 160 in our case, we reduce outliers by tuning min_samples to smaller values compared to min_cluster_size (the default), which reduces core distance, hence maintaining a more robust clustering. Compared to DBSCAN, HDBSCAN provides a more flexible approach for complex astronomical datasets, capturing subtle structures that DBSCAN may overlook.

Results & Discussion

K-Means Clustering

The three candidate values of K were evaluated using internal and external metrics against ground truth labels (Figure 5), supplemented by interpretability and domain knowledge. K=2 achieved the strongest performance (ARI: 0.59, NMI: 0.64), but collapsed galaxies and QSOs into one group, making it physically incomplete. K=3 aligned with the physical prior of the three classes with poor external agreement (ARI: 0.21, NMI: 0.40), with geometric overlap between galaxies and QSOs in feature space reduced cluster separation. K=4 partially resolved this by isolating QSOs into a distinct cluster, at the cost of reduced interpretability against the original three classes.

PCA visualisations revealed that galaxy forms an elongated, non-spherical distribution (Figure 6), violating K-Means' spherical cluster assumption (Turner et al., 2018). These findings motivate the use of methods better suited to non-spherical clusters like GMM and DBSCAN/HDBSCAN.

GMM

K = 3 performed better than K = 10 across all internal and external metrics except cluster purity (Table 1). This is supporting evidence that K = 3 is an appropriate number of components. GMM K = 3 also slightly outperformed the K-means K = 3 (NMI: 0.43 vs 0.39), which is expected. GMM K=3 is chosen for further analysis. The model's max probability assignment indicates high posterior confidence. The concentration of values near 1.0 in the probability assignment distribution (Figure 7) suggests that the model is confident in its assignments. However, this confidence does not translate to strong agreement with ground truth, as indicated by weaker ARI and NMI scores. While galaxies are well separated, stars are absorbed into a quasar-dominated cluster, as seen in the cluster composition tables (Table 2). Overall, the findings suggest a 'confident wrong model' that is capturing structures in feature space that are not separating the labels. This suggested limitation may lie in the feature space rather than the model, with insufficient discriminative information to separate classes.

DBSCAN & HDBSCAN

DBSCAN is not well-suited for this dataset, as it collapses the three-class structure into two clusters (Figure 8a). This happens because DBSCAN uses a single global density threshold (eps). When different classes have different densities, one eps cannot fit all (an eps value suitable for a dense region may not be suitable for a sparse region). HDBSCAN addresses this by modelling local density variations using mutual reachability distance, allowing it to better separate clusters of different densities and identify 3 distinct clusters (Figure 8b), with an optimal min_cluster_size of 160. We further fine-tuned the model by reducing min_samples (default equals min_cluster_size), which initially caused excessive outliers. Lowering it allowed more points to be assigned to clusters (Figure 8c), improving overall class recovery while achieving the best ARI and NMI too (Table 3).

Overall Comparison of Models

K-Means serves as a weak baseline, as its spherical clustering assumption fails to capture the elongated and overlapping structure. GMM offers some improvement by allowing elliptical clusters, but it still struggles to separate the three classes. Moving to density-based methods, HDBSCAN better captures arbitrary shapes and varying densities, resulting in clearer class separation than all prior models (Table 4). This justifies our final choice of HDBSCAN for tackling the problem.

Comparison to Human Performance & Societal Impacts

Models need not outperform humans to be useful; they automate initial classification and reveal data structure. Privacy is not a concern due to public data, though observational bias may affect fairness. These methods support, rather than replace, astronomers by reducing workload and guiding further analysis.

Appendix

AI Usage Declaration

AI was used in this project for writing clarity and polish, specifically to tighten phrasing in the report and suggest figure captions. All analysis, feature engineering decisions, model implementation, hyperparameter tuning, and interpretation of results were conducted independently by the team.

Figures & Tables

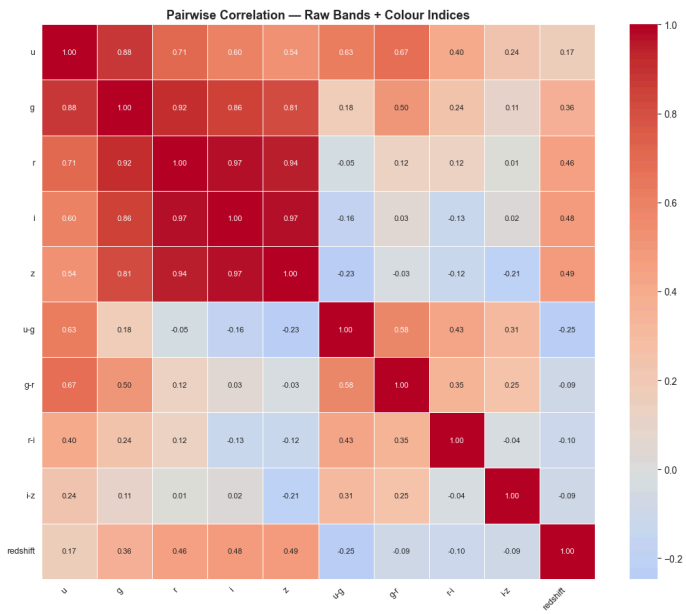


Figure 1: Pairwise correlation heatmap

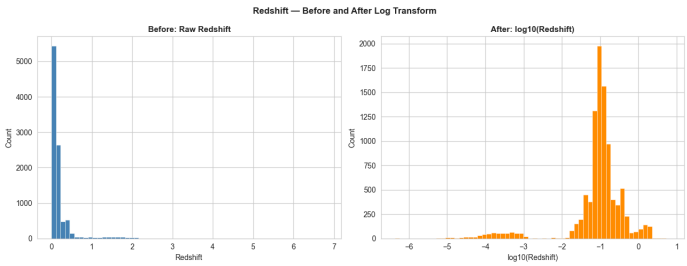


Figure 2: Redshift distribution before and after transformation

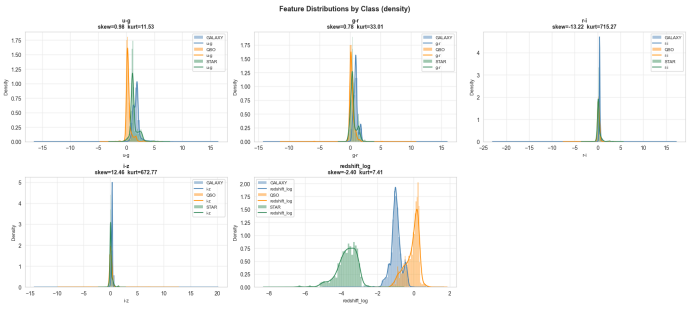


Figure 3: Feature distributions by class before scaling

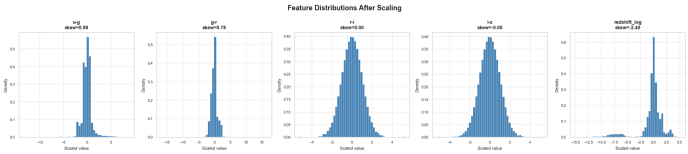


Figure 4: Feature distributions after scaling

k	DB Index	ARI	NMI
2	0.5287	0.5935	0.6355
3	0.8935	0.2062	0.3970
4	0.7788	0.3038	0.5260

K = 2: Cluster x True Class

Cluster	Galaxy	QSO	Star	Class	Purity	Size
1	8412	685	1	Galaxy	92.5%	9098
2	11	0	624	Star	98.3%	635

K = 3: Cluster x True Class

Cluster	Galaxy	QSO	Star	Class	Purity	Size
1	4	0	624	Star	99.4%	624
2	4859	51	1	Galaxy	98.9%	4911
3	3560	634	0	Galaxy	84.9%	4194

K = 4: Cluster x True Class

Cluster	Galaxy	QSO	Star	Class	Purity	Size
1	4622	83	0	Galaxy	98.3%	4745
2	3725	34	1	Galaxy	99.1%	3760
3	5	0	624	Star	99.2%	629
4	31	568	0	QSO	94.8%	599

Figure 5: Evaluation of clustering performance for K = 2, 3, 4 using metrics, with cluster–true class alignment.

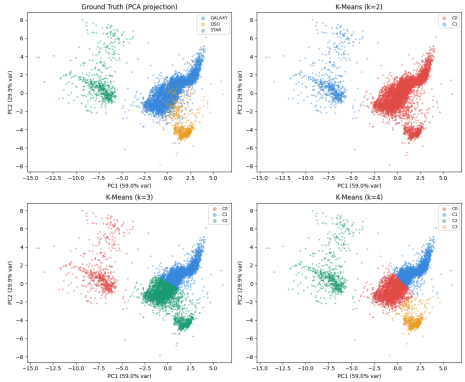


Figure 6: PCA projection comparing ground truth labels with K-means clustering results for K = 2, K = 3 and K = 4

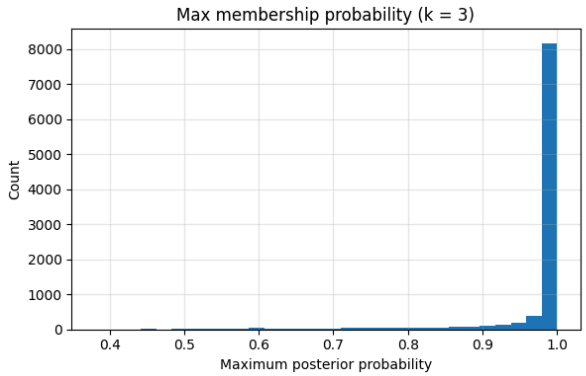


Figure 7: Distribution of maximum posterior membership probabilities for GMM (K=3)

k	BIC	ARI	NMI
3	81,054	0.4619	0.4321
10	57,285	0.0831	0.2988

Table 1: Comparison of clustering performance for K=3 and K=10 using BIC, ARI, and NMI

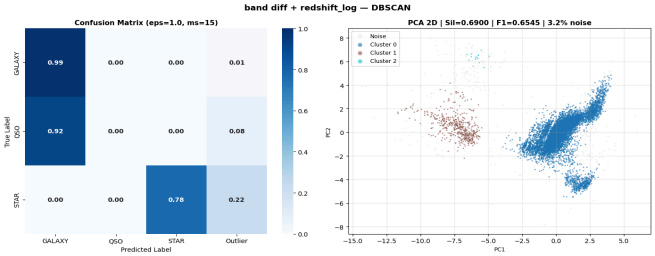
<u>K = 3: Cluster x True Class</u>

Cluster	Galaxy	QSO	Star	Class
1	6890	25	0	Galaxy
2	1150	23	0	Galaxy
3	383	627	625	QSO

Table 2: Cross-tabulation of GMM clusters (K=3) against true classes

Model	DBSCAN (eps: 1.0, min_sample: 15)	HDBSCAN (mc: 160)	HDBSCAN (mc: 160, min_sample: 25)
DBI	0.3673	0.4276	0.4699
ARI	0.5845	0.7752	0.8528
NMI	0.5301	0.6396	0.7210

Table 3: Metric Results of DBSCAN & HDBSCAN



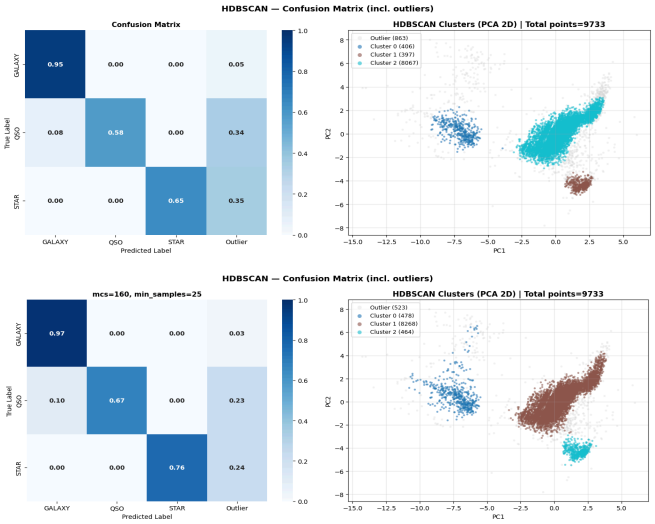


Figure 8 : Clustering Results of DBSCAN & HDBSCAN (Top, Mid, Bottom are respectively Figure 8a, 8b, 8c)

Model	K-Means	GMM	DBSCAN	HDBSCAN
ARI	0.2062	0.4619	0.5845	0.8528
NMI	0.397	0.4321	0.5301	0.7210

Table 4: Clustering Model Performance Comparison across K-Means, GMM, DBSCAN and HDBSCAN (ARI and NMI Scores)

<u>References</u>

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